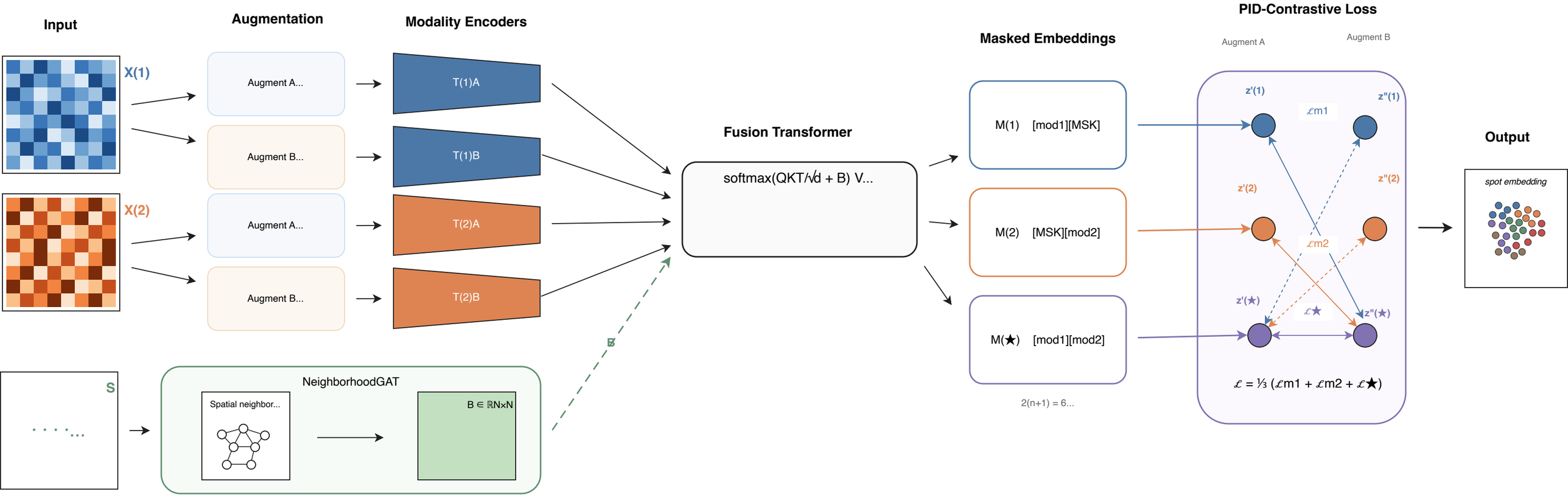

Input
Augmentation
Modality Encoders
X(1)
X(2)
S
Augment A...
Augment B...
Augment A...
Augment B...
T(1)A
T(1)B
T(2)A
T(2)B
NeighborhoodGAT
Spatial neighbor...
B ∈ ℝN×N
B
Fusion Transformer
softmax(QKT/√d + B) V...
Masked Embeddings
M(1) [mod1][MSK]
M(2) [MSK][mod2]
M(★) [mod1][mod2]
2(n+1) = 6...
PID-Contrastive Loss
Augment A
Augment B
z'(1)
z"(1)
ℒm1
z'(2)
z"(2)
ℒm2
z'(★)
z"(★)
ℒ★
ℒ = ⅓ (ℒm1 + ℒm2 + ℒ★)
Output
spot embedding